Introduction to Cloud Computing

Cloud Computing is the **on-demand delivery** of computing services (compute, storage, database, networking, and other services) over the Internet ("the cloud"). with pay as you go pricing

Compute is the processing power of your machine.

There are four core cloud computing services—compute, storage, database, and networking

Ideal Definition of Cloud Computing (As per NIST - National Institute of Standards and Technology):

"Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."

Term	Meaning
Ubiquitous Access	Available anywhere via internet
On-demand	Resources can be accessed when needed
Shared Pool	Resources are shared among users (multi-tenancy)
Rapid Provisioning	Quick scaling up/down
Minimal Management	Less manual work needed for infrastructure

Essential Characteristics of Cloud

As per **NIST** (**National Institute of Standards and Technology**), the essential characteristics are:

1. **On-demand self-service**
    - Users can automatically provision resources (e.g., server time, storage) as needed without human interaction with the service provider.
2. **Broad network access**
    - Services are accessible over the network using standard devices like laptops, smartphones, tablets.
3. **Resource pooling**
    - Provider's (computing) resources are pooled to serve multiple users using multi-tenancy.
4. **Rapid elasticity**
    - Resources can scale out/in quickly to meet demand.

5. **Measured service**
    - Resource usage is monitored, controlled, and billed based on consumption (pay-as-you-go).

Essential Characteristics

Feature	Explanation
On-demand self-service	Users can access resources anytime
Broad network access	Access via standard devices
Resource pooling	Shared resources for multiple users
Rapid elasticity	Resources can scale up/down
Measured service	Pay as you use

Note: -

1. **Multitenancy** (or **multi-tenancy**) is a software architecture concept where a **single instance** of an application **serves multiple customers**, known as **tenants**. Each tenant shares the same software and infrastructure but has **separate and isolated data**.
2. **Resource Pooling** means that a cloud provider (or system) shares **a pool of computing resources**—such as servers, storage, memory, and network bandwidth—across **multiple consumers or tenants**, **dynamically assigning and reassigning** resources as needed.
3. **Location Independence in Cloud Storage:** The customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter).

   e.g:- **Amazon S3**: When you upload a file, S3 automatically stores it in multiple locations within a region. You don't know where exactly—but it's always available.
4. **Pay-as-You-Go (PAYG) Model – Cloud Computing Billing**

   Pay-as-you-go is a **billing model** where customers are **charged based on actual usage** of computing resources rather than a fixed cost or subscription.

   Common Resources Charged in PAYG:

Resource	**Measured in**
Compute (e.g., EC2)	Seconds/minutes/hours
Storage (e.g., S3)	GB per month
Bandwidth	GB transferred

Resource	Measured in
Database queries	Requests, storage, throughput
API Gateway	Number of API calls

Cloud Service Models (SPI Model)

1. **IaaS (Infrastructure as a Service)**
   - Provides virtualized computing resources over the internet.
   - *Examples:* AWS EC2, Google Compute Engine, Azure VMs.
2. **PaaS (Platform as a Service)**
   - Provides a platform allowing users to develop, run, and manage applications.
   - *Examples:* Google App Engine, Heroku, Azure App Services.
3. **SaaS (Software as a Service)**
   - Delivers software applications over the internet on a subscription basis.
   - *Examples:* Gmail, Google Docs, Salesforce, Microsoft 365.

Cloud Deployment Models

1. **Public Cloud**
   - Services offered over the public internet; shared infrastructure.
   - *Example:* AWS, Microsoft Azure, Google Cloud.
2. **Private Cloud**
   - Exclusive to **Shared infrastructure** a single organization; offers greater control and security.

   Note: - Private cloud However, the infrastructure can be maintained by the organization itself, or by the third party that is providing the cloud services. Similarly, the hardware can be located on-site or on the third-party site

   -
3. **Hybrid Cloud**
   - Combines public and private clouds; allows data and apps to move between them.
4. **Community Cloud**
   - for a specific community with shared concerns.

Note: - • Hybrid **Cloud** = A **mix** of cloud types (like public + private) for **flexibility** and **performance**.

• Community **Cloud** = A **shared cloud** for organizations with **similar goals**, like universities, banks, or health systems.

Hybrid Cloud vs Community Cloud

Aspect	Hybrid Cloud	Community Cloud
Definition	A combination of **two or more cloud types** (public, private, or community) integrated together.	A cloud infrastructure **shared by multiple organizations** with common interests or requirements.
Ownership	Owned and managed by **a single organization** or **multiple parties**.	Owned and used **collectively by a group** of similar organizations.
Purpose	To gain benefits of **multiple cloud models** (e.g., security of private + scalability of public).	To **collaborate and share resources** securely among organizations with **common needs**.
Use Case	A company stores sensitive data on a private cloud and runs public apps on a public cloud.	Universities or government departments sharing research data or services.
Examples	Bank uses private cloud for transactions + public for analytics.	Hospitals sharing patient data under strict regulations.

Cloud Service Brokerage (CSB)

Cloud Service Brokers act as intermediaries between cloud service providers and customers. They help in:

- Service aggregation
- Integration
- Customization
- Security enhancement
- Usage monitoring

Example: RightScale, Jamcracker.

Considerations for Building Cloud Infrastructure

1. **Scalability and Elasticity** – Infrastructure must scale with demand.
2. **Security and Compliance** – Identity management, data protection, encryption.
3. **Performance** – Low latency, high throughput networking and storage.
4. **Reliability and Redundancy** – Failover mechanisms, backup, disaster recovery.

5. **Automation and Orchestration** – Tools for provisioning, monitoring, and managing infrastructure.
6. **Cost Management** – Tools to monitor usage and optimize spending.
7. **Multi-tenancy and Isolation** – Ensure secure separation between users.

Note:-

Vertical Scaling (Scale Up) – Add more power to one machine.

Horizontal Scaling (Scale Out) – Add more machines to work together